HARYANA LEET 2026 — FORMULA & QUICK REVISION SHEET

Section A: Basic Sciences | Questions 1 – 25 | 25 Marks

90 Questions · 90 Minutes · +1 / No Negative Marking · English Medium · Online MCQ

a-1 Mathematics (8Q)	a-2 Physics (8Q)	a-3 Chemistry (3Q)	a-4 Comm. Skills (3Q)	a-5 Gen. Awareness (3Q)

[a-1] MATHEMATICS

Arithmetic Progressions (AP)

- *nth term: $a_n = a + (n-1)d$*
- *Sum of n terms: $S_n = n/2 \cdot [2a + (n-1)d]$*
- *Sum of n terms: $S_n = n/2 \cdot (a + l)$, where l = last term*

Geometric Progression (GP)

- *nth term: $a_n = a \cdot r^{n-1}$*
- *Sum: $S_n = a(r^n-1)/(r-1)$, $r \neq 1$*
- *Sum to infinity ($|r|<1$): $S_\infty = a/(1-r)$*

Complex Numbers

- *$i = \sqrt{(-1)}$, $i^2=-1$, $i^3=-i$, $i^4=1$*
- *$|z| = \sqrt{(a^2+b^2)}$, $\arg(z) = \tan^{-1}(b/a)$*
- *De Moivre: $(\cos\theta+i \sin\theta)^n = \cos n\theta + i \sin n\theta$*

Logarithm

- *$\log(mn) = \log m + \log n$*
- *$\log(m/n) = \log m - \log n$*
- *$\log m^n = n \cdot \log m$*
- *Change of base: $\log_b a = \log a / \log b$*

★ $\log_e$ = ln (natural log); $\log_{10}$ = common log

Permutation & Combination

- *${}^nP_r = n! / (n-r)!$*
- *${}^nC_r = n! / [r!(n-r)!]$*
- *${}^nC_0 = {}^nC_n = 1$; ${}^nC_r = {}^nC_{n-r}$*

Binomial Theorem

- *$(a+b)^n = \Sigma_{r=0}^{n}\, {}^nC_r \cdot a^{n-r} \cdot b^r$*
- *General term $T_{r+1} = {}^nC_r \cdot a^{n-r} \cdot b^r$*

✔ Middle term: (n/2+1)th term when n is even

Probability

- *P(A) = Favourable outcomes / Total outcomes*
- *$P(A \cup B) = P(A)+P(B)-P(A \cap B)$*
- *$P(A') = 1 - P(A)$ [Complementary]*
- *$P(A \cap B) = P(A) \cdot P(B)$ [Independent events]*

Matrices & Determinants

- *det[a b; c d] = ad – bc*
- *$A^{-1} = \text{adj}(A) / \det(A)$*

• Singular matrix: det(A) = 0 → no inverse

- *For 2×2: adj[a b;c d] = [d –b; –c a]*

Trigonometry

- *$\sin^2\theta + \cos^2\theta = 1$*
- *$1 + \tan^2\theta = \sec^2\theta$*
- *$1 + \cot^2\theta = \csc^2\theta$*
- *sin(A±B) = sinA cosB ± cosA sinB*
- *cos(A±B) = cosA cosB ■ sinA sinB*

- $\sin 2A = 2 \sin A \cos A$
- $\cos 2A = 2\cos^2 A - 1 = 1 - 2\sin^2 A = \cos^2 A - \sin^2 A$
- $\tan 2A = 2\tan A/(1 - \tan^2 A)$

Straight Line & Circle

- Slope $m = (y_2 - y_1)/(x_2 - x_1) = \tan\theta$
- Line: $y - y_1 = m(x - x_1)$
- Distance between points $= \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$
- Distance from (h,k) to $ax+by+c=0$: $|ah+bk+c|/\sqrt{(a^2+b^2)}$
- Circle std form: $(x-h)^2 + (y-k)^2 = r^2$

Differentiation

- $d/dx(x^n) = nx^{n-1}$
- $d/dx(\sin x) = \cos x$ | $d/dx(\cos x) = -\sin x$
- $d/dx(\tan x) = \sec^2 x$ | $d/dx(e^x) = e^x$
- $d/dx(\ln x) = 1/x$
- Product: $(uv)' = uv' + vu'$
- Quotient: $(u/v)' = (vu' - uv')/v^2$
- Chain rule: $dy/dx = (dy/du)\cdot(du/dx)$
- Maxima: $f'(x)=0$ and $f''(x)<0$ | Minima: $f'(x)=0$ and $f''(x)>0$

Integration

- $\int x^n dx = x^{n+1}/(n+1)+C,\ n \neq -1$
- $\int \sin x\, dx = -\cos x + C$ | $\int \cos x\, dx = \sin x + C$
- $\int e^x dx = e^x + C$ | $\int (1/x) dx = \ln|x| + C$
- $\int \sec^2 x\, dx = \tan x + C$
- Definite: $\int_a^b f(x)dx = F(b) - F(a)$ [Newton-Leibniz]

✔ **Integration by parts: $\int u\, dv = uv - \int v\, du$ (ILATE order)**

ODE — First Order

- Separable: $f(y)dy = g(x)dx \rightarrow$ integrate both sides
- Linear: $dy/dx + P(x)y = Q(x) \rightarrow IF = e^{\int P\, dx}$

Statistics

- Mean x■ $= \Sigma x_i/n$ (ungrouped)
- Mean (grouped) $= \Sigma f_i x_i / \Sigma f_i$
- Standard Deviation $\sigma = \sqrt{[\Sigma(x_i - x■)^2/n]}$
- Mean Deviation $= \Sigma|x_i - x■|/n$
- Median: middle value (odd n) or avg of 2 middle values (even n)
- Mode: value with highest frequency

[a-2] PHYSICS

Units & Dimensions

- SI base units: m, kg, s, A, K, mol, cd
- Dimensional formula of Force: $[MLT^{-2}]$
- Dimensional formula of Energy: $[ML^2T^{-2}]$

✔ **Dimensional analysis: used to check equations & derive formulae**

Motion (1D) & Newton's Laws

- $v = u + at$
- $s = ut + ½at^2$
- $v^2 = u^2 + 2as$
- $F = ma$ (Newton 2nd) | Impulse $= F\Delta t = \Delta p$
- Weight $W = mg$ ($g = 9.8\ m/s^2$)

Work, Energy & Power

- $W = F\cdot d\cdot\cos\theta$
- $KE = ½mv^2$ | $PE = mgh$
- Power $P = W/t = Fv\cdot\cos\theta$

✔ **Conservation: KE+PE = constant (no friction)**

Properties of Matter

- ■ ***Young's Modulus $Y = (F/A) / (\Delta L/L)$ = Stress/Strain***
- ■ ***Surface tension $T = F/L$ (N/m)***
- ■ ***Viscosity (Poiseuille): $Q = \pi r^4 \Delta P/(8\eta L)$***

Waves & SHM

- ■ ***$v = f\lambda$ (wave speed)***
- ■ ***$T = 2\pi\sqrt{(L/g)}$ (simple pendulum)***
- ■ ***$T = 2\pi\sqrt{(m/k)}$ (spring-mass)***
- ■ ***Energy in SHM $\propto A^2$ (A = amplitude)***

Rotational Motion

- ■ ***Torque $\tau = I\cdot\alpha$ | Angular momentum $L = I\omega$***
- ■ ***$KE_{rot} = ½I\omega^2$***
- ■ ***$I_{ring}=mr^2$ | $I_{disc}=½mr^2$ | $I_{sphere}=2mr^2/5$***

Heat & Temperature

- ■ ***$Q = mc\Delta T$ (c = specific heat)***
- ■ ***Linear expansion: $\Delta L = L\alpha\Delta T$***
- ■ ***$PV = nRT$ (Ideal Gas Law)***
- ■ ***Fourier conduction: $Q/t = kA(T_1-T_2)/d$***

Geometric Optics

- ■ ***Mirror formula: $1/f = 1/v + 1/u$ ($f=R/2$)***
- ■ ***Snell's law: $n_1\sin\theta_1 = n_2\sin\theta_2$***
- ■ ***Lens formula: $1/f = 1/v - 1/u$***
- ■ ***Magnification $m = v/u$ (mirror & lens)***
- ■ ***Lensmaker: $1/f = (n-1)[1/R_1-1/R_2]$***

Electrostatics & Capacitors

- ■ ***Coulomb: $F = kq_1q_2/r^2$, $k=9\times10^9$ Nm^2/C^2***
- ■ ***$E = kq/r^2$ | $V = kq/r$***
- ■ ***$C = Q/V$ | Parallel plates: $C = \varepsilon_0 A/d$***
- ■ ***Energy $= ½CV^2 = Q^2/2C$***
- ■ ***C_{series}: $1/C=1/C_1+1/C_2$ | $C_{parallel}$: $C=C_1+C_2$***

Modern Physics — Laser & Superconductivity

- LASER: Light Amplification by Stimulated Emission of Radiation
- Properties: monochromatic, coherent, directional, high intensity
- Superconductivity: zero electrical resistance below critical temp T_c
- Conventional energy: coal, oil, gas | Non-conventional: solar, wind, tidal, nuclear

[a-3] CHEMISTRY

Water Chemistry

- Hard water: contains dissolved Ca^{2+}, Mg^{2+} salts → doesn't lather with soap
- Temporary hardness: due to $Ca(HCO_3)_2$, $Mg(HCO_3)_2$ → removed by boiling/lime
- Permanent hardness: due to $CaSO_4$, $MgCl_2$ etc → removed by washing soda or ion exchange
- ■ ***Ion Exchange: Na-zeolite + $CaCl_2$ → Ca-zeolite + 2NaCl***
- ■ ***$pH = -\log[H^+]$ | pH<7 Acid | pH=7 Neutral | pH>7 Base***
- ■ ***$pOH = -\log[OH^-]$ | pH + pOH = 14***

Acidity, Basicity & Equivalent Weight

- Acidity of base: no. of OH^- ions it gives per formula unit
- Basicity of acid: no. of H^+ ions it gives per formula unit
- ■ ***Equivalent weight = Molecular weight / Valency (acidity/basicity)***

Electronic Configuration (Z up to 30)

- Fill order: 1s → 2s → 2p → 3s → 3p → 4s → 3d
- Fe(26): [Ar] $3d^6$ $4s^2$ | Cu(29): [Ar] $3d^{10}$ $4s^1$ | Zn(30): [Ar] $3d^{10}$ $4s^2$

✔ **Cr(24) and Cu(29) are exceptions due to extra stability of half/full d-orbital**

[a-4] COMMUNICATION SKILLS

Voice (Active ↔ Passive)

- Active: Subject + V + Object → "Ram eats mango"
- Passive: Object + is/was + V3 + by + Subject → "Mango is eaten by Ram"
- Tense change: eat→is eaten, ate→was eaten, will eat→will be eaten

Narration (Direct ↔ Indirect)

- Say/says → tell/tells | Said → told
- Tense shifts back: is→was, am→was, will→would, can→could, have→had
- Pronouns: I→he/she, we→they, you→he/she/they

Tenses (Quick Reference)

- Simple Present: V1/V1+s | Simple Past: V2 | Simple Future: will+V1
- Present Continuous: is/am/are+V-ing | Past Continuous: was/were+V-ing
- Present Perfect: has/have+V3 | Past Perfect: had+V3

Prepositions & Grammar

- at (point), in (enclosed space/time), on (surface/day), by (agent/means)
- since (point of time), for (duration), between (2), among (3+)
- Correction tip: Subject-verb agreement — singular subject → singular verb

[a-5] GENERAL AWARENESS — HARYANA & LEET

Haryana — Key Facts

- Formed: 1 November 1966 | Capital: Chandigarh | Districts: 22
- Area: 44,212 km² | Language: Hindi | High Court: Chandigarh
- Borders: Punjab (N/W), HP (N/NE), Uttarakhand & UP (E), Rajasthan (S), Delhi (SE)
- Rivers: Ghaggar, Yamuna, Saraswati (seasonal)

LEET 2026 — Exam Facts

- Conducting body: HSTES — Haryana State Technical Education Society, Panchkula
- Admission: 2nd Year (3rd Semester) B.E./B.Tech. Lateral Entry
- Eligibility: 3-yr Diploma in Engg./Tech. ≥45% (40% reserved) OR B.Sc.+12th Math ≥45%
- KM Relaxation: upto 10% in cut-off for Kashmiri Migrants
- Mode: Online MCQ | Duration: 90 min | Total: 90 Qs | Marks: 90 | No negative marking
- Result: inter-se merit list on hstes.org.in | Then online counselling

Counselling Process

- 1. Online Registration → 2. Document Verification → 3. Choice Filling → 4. Locking → 5. Seat Allotment
- 15% seats: Rest of Haryana Quota | 2 online counselling rounds
- After 2nd round: RoHC seats merged into General Category

Technical Education in Haryana

- HSBTE: Haryana State Board of Technical Education — governs diploma programs
- Polytechnics: Govt. + Private (check hstes.org.in for latest count & intake)

★ LEET 2026 official notification: check techeduhry.gov.in / hstes.org.in regularly